25901612 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCBTG100 - APPLIED BIOTECHNOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

One Word Answer

Answer any **10** questions

Each question carries **2** marks

1. What is a potential challenge of implementing personalized medicine? [U] / [CO1]
2. Explain the mechanism of metal recovery in biohydrometallurgy. [U] / [CO2]
3. What is the importance of CRISPR-Cas9 technology? [K] / [CO4]
4. List out the microorganisms used for SCP production. [U] / [CO2]
5. Define transgenic animals? [K] / [CO3]
6. Distinguish between biofuels and fossil fuels in terms of sustainability. [U] / [CO1]
7. Memorize xenograft. [K] / [CO4]
8. What is the primary goal of waste management? [U] / [CO2]
9. Define GMO? [K] / [CO3]
10. What is ELSI? [K] / [CO4]

11. What is the major tools in recombinant DNA technology. [K] / [CO4]

12. Define bio-entrepreneurship [U] / [CO3]

[2x10 = 20]

Part B

Short Essay

Answer any **5** questions

Each question carries **4** marks

13. What are the ethical considerations for emerging Biotechnology? [K] / [CO1]

14. Discuss the objectives of Human Genome Project. [U] / [CO4]

15. What is DBT? Mention it's role. [U] / [CO3]

16. Discuss how biotechnology techniques can be used in forensic science. [U] / [CO4]

17. Clarify the concept of selective breeding through Artificial insemination. [U] / [CO3]

18. Elucidate the main goals of using nanobiotechnology in diagnosis and treatment of diseases? [U] / [CO4]

19. Summarize the importance of enzyme-based processes in improving wine quality and production efficiency. [K] / [CO2]

[4x5 = 20]

Part C

Essay

Answer any **1** questions

Each question carries **10** marks

20. Discuss the role of Golden Rice in addressing vitamin A deficiency and its potential impact on global health. [U] / [CO3]

21. How do bioreactors help in scaling up antibiotic production? [U] / [CO4]

[10x1 = 10]